

DATAOPS & MLOPS

5-UNIT COURSE SYLLABUS

Build • Automate • Deploy • Monitor Modern Data & Machine Learning Workflows

Industry-oriented curriculum covering DataOps foundations, automated data pipelines, MLOps lifecycle management, production deployment and monitoring, cloud MLOps, governance, and responsible AI.

Course Objectives

- Understand DataOps and MLOps principles for modern data-driven organizations.
- Build automated data pipelines and machine learning workflows.
- Implement version control, CI/CD, and model deployment strategies.
- Monitor, maintain, and govern machine learning systems in production.
- Gain hands-on experience with industry-standard tools and platforms.

5-Unit Curriculum at a Glance

Unit	Core Area	Focus
Unit 1	Introduction to DataOps & MLOps	Foundations, lifecycles, roles, collaboration and modern platforms
Unit 2	DataOps Foundations & Pipeline Automation	Ingestion, quality, orchestration, versioning, lineage and Airflow
Unit 3	MLOps Workflow & Model Lifecycle Management	Experiments, features, versioning, packaging, Docker, CI/CD and MLflow
Unit 4	Model Deployment & Monitoring	Serving, APIs, deployment strategies, monitoring, drift, alerts and retraining
Unit 5	Cloud MLOps & Governance	Cloud architecture, governance, Responsible AI, security, privacy and sustainability

UNIT 1 — Introduction to DataOps and MLOps

Unit Summary: Establishes the foundation for understanding how data engineering, software operations, and machine learning operations work together.

Key Topics

- Evolution of Data Science Operations
- DataOps vs DevOps vs MLOps
- Data Lifecycle Management
- Challenges in Traditional Data Science Workflows
- Agile and Lean Principles in Data Operations
- MLOps Fundamentals
- Machine Learning Lifecycle
- Roles and Responsibilities in DataOps Teams
- Collaboration between Data Engineers, Data Scientists, and ML Engineers
- Overview of Modern Data and ML Platforms

Learning Outcome: Understand the purpose, lifecycle, roles, and operational principles of DataOps and MLOps.

UNIT 2 — DataOps Foundations and Pipeline Automation

Unit Summary: Focuses on reliable and repeatable data workflows through ingestion, quality, orchestration, versioning, lineage, and reproducibility.

Key Topics

- Data Ingestion Techniques
- ETL vs ELT Processes
- Data Quality Management
- Data Validation and Profiling
- Metadata Management
- Workflow Orchestration
- Pipeline Scheduling
- Data Versioning Concepts
- Data Lineage
- Reproducibility in Data Pipelines
- Apache Airflow
- DVC (Data Version Control)
- Great Expectations

Learning Outcome: Design and implement automated data pipelines incorporating ingestion, validation, versioning, and workflow orchestration.

UNIT 3 — MLOps Workflow and Model Lifecycle Management

Unit Summary: Moves from individual ML experiments toward reproducible, versioned, containerized, and continuously delivered ML workflows.

Key Topics

- ML Experiment Tracking
- Feature Engineering Pipelines
- Feature Stores
- Hyperparameter Tuning
- Model Versioning
- Model Packaging and Serialization
- Containerization using Docker
- CI/CD Concepts for ML
- Automated Testing for ML Models
- Continuous Training
- Continuous Delivery
- MLflow
- Docker
- GitHub Actions

Learning Outcome: Develop and manage ML workflows using experiment tracking, model versioning, containerization, and CI/CD practices.

UNIT 4 — Model Deployment and Monitoring

Unit Summary: Covers production model serving and reliability through APIs, monitoring, logging, alerting, drift detection, and retraining.

Key Topics

- Model Serving Architectures
- REST APIs for ML Models
- Batch vs Real-Time Inference
- Edge vs Cloud Deployment
- Deployment Strategies
- Monitoring ML Systems
- Concept Drift and Data Drift
- Performance Monitoring
- Logging and Alerting
- Retraining Strategies
- FastAPI
- Flask
- Prometheus
- Grafana
- Evidently AI

Learning Outcome: Deploy, monitor, and maintain machine learning models in production environments using appropriate tools and frameworks.

UNIT 5 — Cloud MLOps and Governance

Unit Summary: Introduces cloud-based MLOps architecture plus governance, ethical, security, privacy, and sustainability considerations.

Key Topics

- Cloud MLOps Architecture
- Data, experimentation, model-management, deployment and monitoring layers
- Benefits: faster deployment, elastic scalability, cost optimization, governance and compliance
- Google Vertex AI
- Amazon SageMaker
- Microsoft Azure ML
- Databricks ML
- ML Governance
- Responsible AI
- Explainable AI
- Security and Privacy in MLOps
- Sustainability in DataOps and MLOps
- Netflix Recommendation System case study
- Uber Michelangelo Platform case study
- Google Vertex AI case study
- Azure ML Studio case study

Learning Outcome: Evaluate cloud-based MLOps solutions, governance frameworks, ethical considerations, and sustainability aspects of AI systems.

Technology Stack Covered

- Apache Airflow — workflow orchestration and pipeline scheduling
- DVC — data versioning
- Great Expectations — data quality and validation
- MLflow — experiment tracking and model lifecycle management
- Docker — containerization
- GitHub Actions — CI/CD automation
- FastAPI / Flask — model-serving APIs
- Prometheus — metrics collection
- Grafana — monitoring dashboards
- Evidently AI — data drift, concept drift and model-quality monitoring
- Google Vertex AI, Amazon SageMaker, Microsoft Azure ML and Databricks ML — cloud MLOps platforms

Course Outcomes

- Explain the concepts, principles, and lifecycle of DataOps and MLOps.
- Design and implement automated data pipelines using ingestion, validation, versioning, and orchestration.
- Develop and manage ML workflows including experiment tracking, model versioning, containerization, and CI/CD.
- Deploy, monitor, and maintain ML models in production.
- Evaluate cloud MLOps solutions, governance, ethical considerations, and sustainability.

Delivery & Customization

The source academic structure allocates 9 hours to each of the five modules (45 hours total). For college, university, and corporate engagements, the same five-unit curriculum can be delivered in a customized duration based on institutional objectives, learner background, practical depth, and project requirements.

Recommended Learning Approach

- Concept explanation followed by demonstrations and practical exercises.
- Hands-on work with relevant tools and platforms.
- Progressive movement from data pipelines to ML lifecycle management and production monitoring.
- Architecture discussions and industry case studies.
- Customized practical depth and project scope for college, university, or corporate audiences.

Reference Books

- Practical MLOps
- Introducing MLOps
- DataOps Fundamentals
- Machine Learning Engineering with Python